PCA and LDA

CS21206: Foundations of AI and ML

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Mar 26 and 27, 2026

Agenda

- § Understanding the concept of dimensionality reduction
- § Principal Component Analysis as maximizing variance in a subspace
- § Linear Discriminant Analysis as minimizing variance in a subspace

Resources

- Lecture 1 of Winter 2017 offering of Unsupervised Learning (STAT 841) at the University of Waterloo by Prof. Ali Ghodsi.
- Lecture 4 of Fall 2015 offering of STAT 441/841 at the University of Waterloo by Prof. Ali Ghodsi.

Matrix Operations

- **Addition:** Matrices can be added as long as they have the same shape, by adding their corresponding elements. $C = A + B$, where $C_{i,j} = A_{i,j} + B_{i,j}$.
- **Multiplication:** In order for the product of the two matrices A and B to be defined, A must have the same number of columns as that of the rows of B. If A is of shape $m \times n$ and B is of shape $n \times p$ then C is of shape $m \times p$, the product operation $C = AB$ is defined by $C_{i,j} = \sum_k A_{i,k}B_{k,j}$.

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- **Elementwise or Hadamard Product:** It's a matrix containing the product of the individual elements. It is denoted as $A \odot B$.
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- **Commutativity:** Matrix product is not commutative ($AB \neq BA$ does not always hold). However, the dot product between two vectors is commutative, i.e., $\boldsymbol{x}^T \boldsymbol{y} = \boldsymbol{y}^T \boldsymbol{x}$.

Eigenvalues and Eigenvectors

- Suppose A is a matrix. Does there exist any vector $\boldsymbol{u}$ for A so that the operation $A\boldsymbol{u}$ gives a vector which is nothing but a stretched (and not rotated) version of the vector $\boldsymbol{u}$?
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§ For non-trivial solution $\det(\lambda I - A) = |\lambda I - A| = 0$.

§ If $A \in \mathbb{R}^{d\times d}$, then $|\lambda I - A| = 0$ will be a d^{th} order equation. That means you can have d solutions of λ - such λ's are called eigenvalues (real or complex conjugate).

§ The corresponding vector $\boldsymbol{u}$'s are the eigenvectors. Remember that eigenvectors are not unique. This is because if $\boldsymbol{u}$ is an eigenvector, then $\alpha\boldsymbol{u}$ is also the same eigenvector (as it satisfies $A(\alpha\boldsymbol{u}) = \lambda(\alpha\boldsymbol{u})$). So, we are satisfied with the direction of the eigenvectors only.

Standard Results on Eigenvalues and Eigenvectors

- If $\lambda_1, \lambda_2, \ldots, \lambda_d$ are eigenvalues of A, then for any positive integer m, $\lambda_1^m, \lambda_2^m, \ldots, \lambda_d^m$ are eigenvalues of A^m. That means, if you have eigenvalues of A, you don't have to compute the eigenvalues of A^m.
- If A is a non-singular or invertible matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_d$, then $\lambda_1^{-1}, \lambda_2^{-1}, \ldots, \lambda_d^{-1}$ are eigenvalues of A^{-1}.
- For triangular matrix (upper, lower or diagonal), the eigenvalues are the diagonal elements itself.
- If a square matrix $A \in \mathbb{R}^{d \times d}$ is symmetric then it has d eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_d$ that are all real-valued. The set of d corresponding eigenvectors $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_d$ form an orthonormal basis of $\mathbb{R}^d$ *i.e.*,

$$\boldsymbol{u}_i^T \boldsymbol{u}_j = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases}$$

- The reverse is also true - if a square matrix $A \in \mathbb{R}^{d \times d}$ has d real eigenvalues and d real orthogonal eigenvectors, then the matrix is symmetric.

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- Then we can write the following:

$$A \underbrace{\begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_d \end{bmatrix}}_{U} = \underbrace{\begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_d \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_d \end{bmatrix}}_{\Lambda}$$

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- $AU = U\Lambda$ *i.e.*, $A = U\Lambda U^{-1}$
- Let us see what would be $U^T U$ for a square and symmetric matrix A? $(U^T U)_{ij} = \boldsymbol{u}_i^T \boldsymbol{u}_j$ which is 1 only when $i = j$.
- That means $U^T U$ is an identity matrix which in turn means, $U^{-1} = U^T$. So, in such a case, $A = U\Lambda U^T$

Singular Value Decomposition

- Suppose $A \in \mathbb{R}^{m \times n}$ with $rank(A) = r$. Then A can be factored as,

$$A = U \Sigma V^T$$

- where $U \in \mathbb{R}^{m \times r}$ has orthogonal columns and satisfies $U^T U = I$,
- $V \in \mathbb{R}^{n \times r}$ has orthogonal columns and satisfies $V^T V = I$,
- $\Sigma = diag(\sigma_1, \sigma_2, \cdots, \sigma_r)$ is a $r \times r$ matrix with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$
- The factorization is called the *singular value decomposition* (SVD) of A.
- The columns of U are called *left singular vectors* of A, the columns of V are *right singular vectors*, and the numbers σ_i are the *singular values*.

SVD and Eigenvalue Decomposition

- SVD of a matrix A is closely related to the eigenvalue decomposition of the matrix A^TA which is symmetric and non-negative definite.

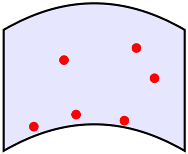
$$A^TA = (U\Sigma V^T)^T(U\Sigma V^T) = V\Sigma^T U^T U\Sigma V^T$$
$$= V\Sigma^2 V^T = \begin{bmatrix} V & \tilde{V} \end{bmatrix} \begin{bmatrix} \Sigma^2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V^T \\ \tilde{V}^T \end{bmatrix}$$

  where $\tilde{V}$ is any matrix such that $\begin{bmatrix} V & \tilde{V} \end{bmatrix}$ is orthogonal.
- The last expression is the eigenvalue decomposition of A^TA.
- So the nonzero eigenvalues of A^TA are squares of the singular values of A. Also the associated eigenvectors of A^TA are the right singular vectors (V) of A.
- Similarly, it can be shown that, the nonzero eigenvalues of AA^T are squares of the singular values of A. Also the associated eigenvectors of AA^T are the left singular vectors (U) of A.

Intuition: Data Residing on Manifold

The Concept: High-dimensional data often resides on a much lower-dimensional *manifold*.

- Imagine dots on a flat sheet of paper.
- The paper is a **2D space** living in a **3D world**.
- Even if we crumple or roll the paper, the "intrinsic" distance between points along the surface stays the same.
- **Manifold:** A space that locally looks like flat Euclidean space but may be globally curved.



2D Manifold in 3D Space

Dimensionality Reduction

Dimensionality Reduction is the act of finding the coordinates of this hidden structure.

§ **Manifold (Non-Linear):** A "curved" surface.

- Example: The surface of the earth (intrinsically 2D, but curved in 3D).
- Requires algorithms like ISOMAP or t-SNE (not covered in this course).

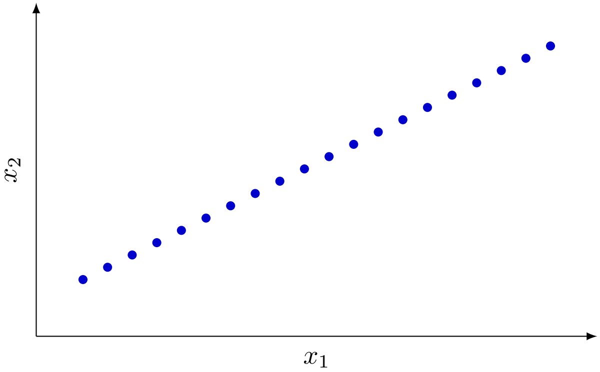
§ **Subspace (Linear):** A "flat" manifold.

- Example: A plane through the origin.
- **PCA** and **LDA** find the best linear subspaces.

The Big Question

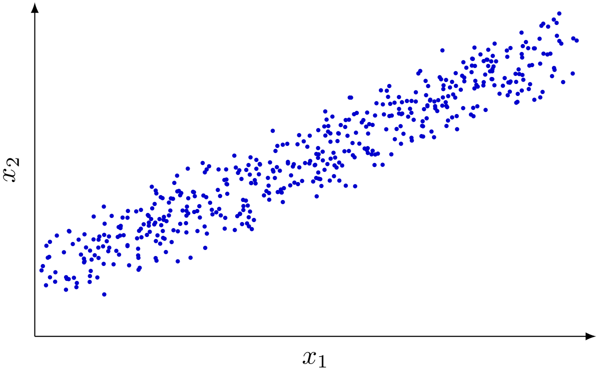
Can we represent data points originally in a high dimension using only a few "important" dimensions?

PCA



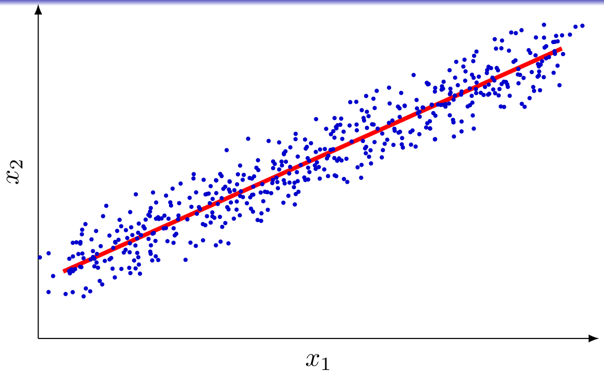
- I can describe the points with a single dimensional coordinates instead of the original 2-dimensional space they reside.
- This is a subspace which fully describes the data

PCA



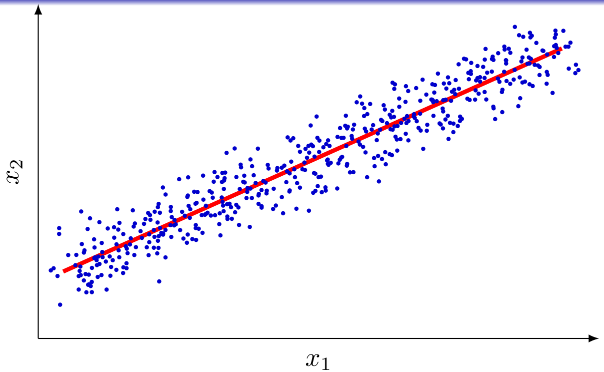
§ What about this case?

PCA



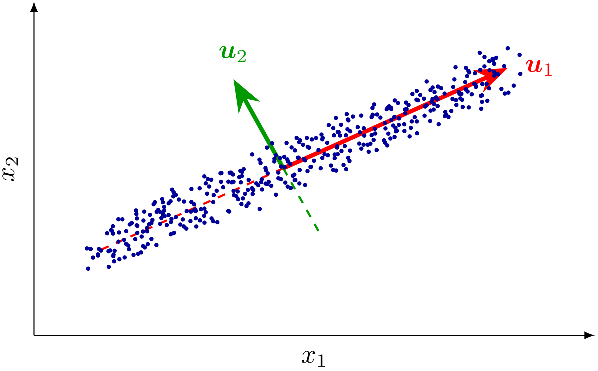
- We will still be able to compute the subspace (straight line here) and we will have to project the points on to the subspace.

PCA



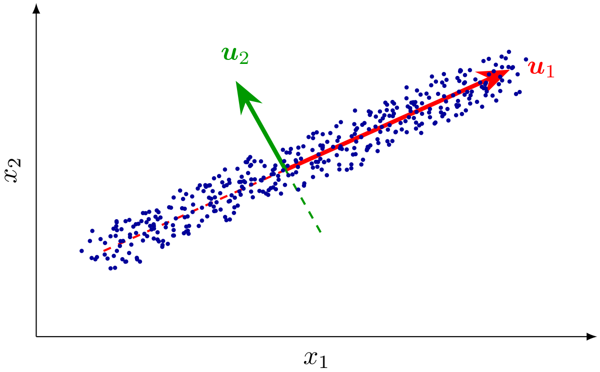
- We will still be able to compute the subspace (straight line here) and we will have to project the points on to the subspace.
- Principal Component Analysis (PCA) allows us to find such subspaces.
- PCA is a method of dimensionality reduction/feature extraction that transforms the data from a d-dimensional space into a new coordinate system of dimension p, where $p \leq d$ (the worst case would be to have $p = d$).

PCA



- The direction $\boldsymbol{u}_1$ is called the *principal component* of the subspace.
- There are other principal components (*e.g.*, $\boldsymbol{u}_2$) that are orthogonal to the first principal component.

PCA



- The direction $\boldsymbol{u}_1$ is called the *principal component* of the subspace.
- There are other principal components (*e.g.*, $\boldsymbol{u}_2$) that are orthogonal to the first principal component.
- One view of PCA is - rotating the coordinate system such that the data resides in a lower dimensional subspace.

PCA

§ Another way of looking at PCA is: Among all possible vectors in the d-dimensional (here $d = 2$) space, we are looking for the vector $\boldsymbol{u}_1$ such that when the data is projected in the direction of $\boldsymbol{u}_1$, the variation of the projections is maximum.

§ We will derive PCA using this view. But before that some important information.

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- We will derive PCA using this view. But before that some important information.
- In PCA, the goal is to preserve as much of the variance in the original data as possible in the new coordinate systems.
- Given d-dimensional data, the hope is that the data points will lie mainly in a linear subspace of dimension lower than d.
- In practice, the data will usually not lie precisely in some lower dimensional subspace.
- The new variables that form a new coordinate system are called principal components (PCs).

PCA

- PCs denoted by $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_d$ form a basis for the data.
- Normally, all PCs are not used, rather a subset of PCs $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_p$ $[p < d]$ are used.
- The most common definition of PCA, due to Hotelling is that, for a given set of data vectors $\boldsymbol{x}_i, i \in 1, \cdots, n$, the p principal axes are those orthonormal axes onto which the variance retained under projection is maximal.

PCA: Derivation

- Suppose we have n data points each of which are d-dimensional.

$$X = \left[\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n\right]_{d \times n}, \ \boldsymbol{x}_i \in \mathbb{R}^d \quad \forall i = 1, \cdots n$$

- The projection of all these data points on vector $\boldsymbol{u}_1 \in \mathbb{R}^d$ is $\boldsymbol{u}_1^T X$.
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- So, the optimization problem is:

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- However, this is a quadratic function and a quadratic's maximum value is ∞.
- So, we need a constraint. And a natural constraint is fix the length of $\boldsymbol{u}_1$ to 1 (As ultimately, we are interested in the direction of the principal component only).

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$$\text{sub to: } \boldsymbol{u}_1^T \boldsymbol{u}_1 = 1$$

- The Lagrangian: $L(\boldsymbol{u}_1, \lambda) = \boldsymbol{u}_1^T S \boldsymbol{u}_1 - \lambda(\boldsymbol{u}_1^T \boldsymbol{u}_1 - 1)$

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- But, S being a covariance matrix its positive semidefinite and can have even d eigenvalues and eigenvectors. Which one of them actually maximizes the objective?
- $\boldsymbol{u}_1^T S \boldsymbol{u}_1 = \boldsymbol{u}_1^T \lambda \boldsymbol{u}_1$ [since, $S\boldsymbol{u}_1 = \lambda \boldsymbol{u}_1$]. *i.e.*, $\boldsymbol{u}_1^T S \boldsymbol{u}_1 = \lambda \boldsymbol{u}_1^T \boldsymbol{u}_1 = \lambda$
- The highest value of the objective happens when the component is chosen along the eigenvector corresponding to the maximum eigenvalue.

PCA via SVD

- Recall, $A = U\Sigma V^T$ and,
- The eigenvectors of A^TA are the right singular vectors (V) of A.
- The eigenvectors of AA^T are the left singular vectors (U) of A.
- This means: U of the SVD (of A) contains the eigenvectors of AA^T

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- The eigenvectors of A^TA are the right singular vectors (V) of A.
- The eigenvectors of AA^T are the left singular vectors (U) of A.
- This means: U of the SVD (of A) contains the eigenvectors of AA^T
- For us, $X = \left[\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n\right]$
- Let, $\tilde{X} = \left[\boldsymbol{x}_1 - \bar{\boldsymbol{x}} \quad \boldsymbol{x}_2 - \bar{\boldsymbol{x}} \quad \cdots \quad \boldsymbol{x}_n - \bar{\boldsymbol{x}}\right]$, where $\bar{\boldsymbol{x}} = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i$
- Now $\tilde{X}\tilde{X}^T$ is the covariance matrix of X
- Let us collect the important informations in the next slide.

PCA via SVD

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- So, if we take the SVD of $\tilde{X} = U\Sigma V^T$, then columns of U will be the eigenvectors of $\tilde{X}\tilde{X}^T$.
- And, eigenvectors of $\tilde{X}\tilde{X}^T$ are the principal components of X (as $\tilde{X}\tilde{X}^T$ is the "sample" covariance of X).

PCA via SVD

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- So, if we take the SVD of $\tilde{X} = U\Sigma V^T$, then columns of U will be the eigenvectors of $\tilde{X}\tilde{X}^T$.
- And, eigenvectors of $\tilde{X}\tilde{X}^T$ are the principal components of X (as $\tilde{X}\tilde{X}^T$ is the "sample" covariance of X).
- So, instead of computing the sample covariance and then taking its eigendecomposition, Principal Components can be easily obtained by taking SVD of $\tilde{X}$.
- Computing $\tilde{X}\tilde{X}^T$ and taking eigendecomposition involves squaring the data, which can lead to a loss of precision (low condition number). SVD operates directly on $\tilde{X}$.
- For high-dimensional data where $d \gg n$, SVD algorithms are often more computationally efficient than explicitly forming the $d \times d$ covariance matrix.

Linear Discriminant Analysis

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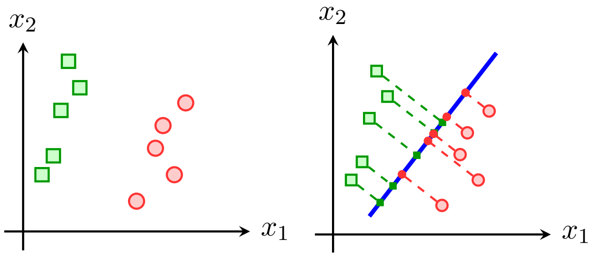
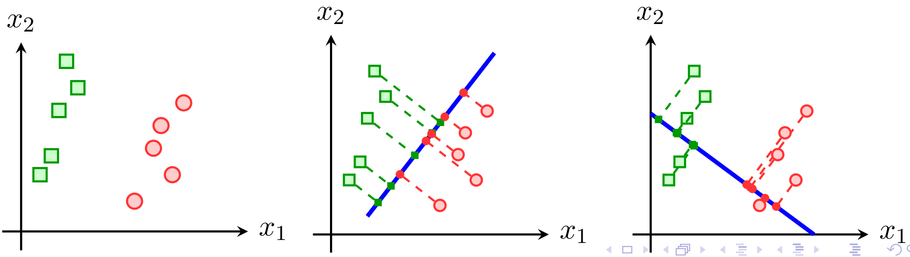
- For classification, this would be a terrible projection, because all labels get evenly mixed and we destroy the useful information.



Linear Discriminant Analysis

- We studied PCA as a technique to reduce dimensionality
- It searches for directions in the data that have largest variance and subsequently projects the data onto it.
- PCA is unsupervised (does not use class information of data).
- For classification, this would be a terrible projection, because all labels get evenly mixed and we destroy the useful information.
- A much more useful projection would be orthogonal to this direction, *i.e.*, in the direction of least overall variance.
- This would perfectly separate the data.



Linear Discriminant Analysis

- How do we utilize the label information in finding informative projections?
- The major "Philosophy" in LDA is: reducing data variation in the same class and increasing the separation between classes.

Linear Discriminant Analysis

- How do we utilize the label information in finding informative projections?
- The major "Philosophy" in LDA is: reducing data variation in the same class and increasing the separation between classes.
- We project the samples onto a line (that reduces the dimension to a single dimension) parameterized by $\boldsymbol{w}$ as $y = \boldsymbol{w}^T\boldsymbol{x}$
- Then find the $\boldsymbol{w}^*$ that maximizes the ratio of "between-class" variance and "within-class" variance.

Linear Discriminant Analysis

§ Assume, we have a set of d-dimensional samples $X = \left[\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n\right]$, N_1 of which belong to class $\mathcal{C}_1$, and N_2 of which belong to class $\mathcal{C}_2$.

§ The mean vectors of the two classes in the d-dimensional space are:

$$\boldsymbol{\mu}_1 = \frac{1}{N_1} \sum_{i \in \mathcal{C}_1} \boldsymbol{x}_i \qquad \boldsymbol{\mu}_2 = \frac{1}{N_2} \sum_{i \in \mathcal{C}_2} \boldsymbol{x}_i \tag{1}$$

Linear Discriminant Analysis

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- The mean vectors of the projections of the two classes are:

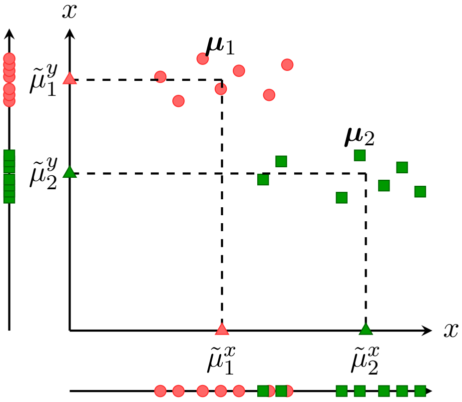
$$\begin{aligned}
\tilde{\mu}_1 &= \frac{1}{N_1}\sum_{i\in\mathcal{C}_1}\boldsymbol{w}^T\boldsymbol{x}_i & \qquad \tilde{\mu}_2 &= \frac{1}{N_2}\sum_{i\in\mathcal{C}_2}\boldsymbol{w}^T\boldsymbol{x}_i \\
&= \boldsymbol{w}^T\left(\frac{1}{N_1}\sum_{i\in\mathcal{C}_1}\boldsymbol{x}_i\right) & &= \boldsymbol{w}^T(\frac{1}{N_2}\sum_{i\in\mathcal{C}_2}\boldsymbol{x}_i) \\
&= \boldsymbol{w}^T\boldsymbol{\mu}_1 & &= \boldsymbol{w}^T\boldsymbol{\mu}_2
\end{aligned} \tag{2}$$

How about separation between means

- $|\tilde{\mu}_1 - \tilde{\mu}_2|$ seems like a good measure.

How about separation between means

- $|\tilde{\mu}_1 - \tilde{\mu}_2|$ seems like a good measure.
- The vertical projection is better than the horizontal projection for class separability.
- however, $|\tilde{\mu}_1^x - \tilde{\mu}_2^x| > |\tilde{\mu}_1^y - \tilde{\mu}_2^y|$



Variance is not considered

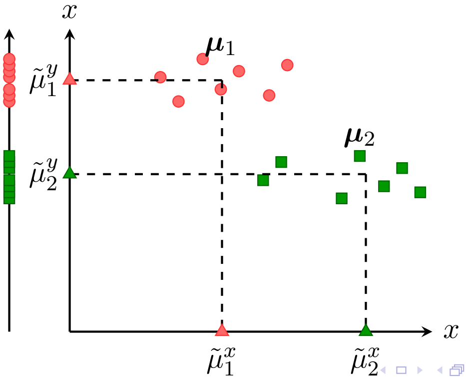
- The problem is that variance of the classes is not considered.

Variance is not considered

- The problem is that variance of the classes is not considered.
- With projection $y_i = \boldsymbol{w}^T \boldsymbol{x}_i$, define 'scatter' of a class in projection space as:

$$\tilde{S}_1^2 = \sum_{i \in \mathcal{C}_1} (y_i - \tilde{\mu}_1)^2 \qquad \tilde{S}_2 = \sum_{i \in \mathcal{C}_2} (y_i - \tilde{\mu}_2)^2 \tag{3}$$

- scatter is proportional to variance, the spread of data around the mean.



Normalize by Variance

- We need to normalize $|\tilde{\mu}_1 - \tilde{\mu}_2|$ by the scatter of the data in the projection space.
- $J(\boldsymbol{w}) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2}$
- If we maximize $J(\boldsymbol{w})$, the numerator will push the projected means as far apart as possible from each other.
- The denominator will try to minimize the scatter of both the classes as low as possible. This means, samples of individual classes will cluster around the projected mean.

Linear Discriminant Analysis

$$J(\boldsymbol{w}) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2} \tag{4}$$

§ Let us play with the numerator.

$$\begin{aligned}(\tilde{\mu}_1 - \tilde{\mu}_2)^2 &= (\tilde{\mu}_1 - \tilde{\mu}_2)(\tilde{\mu}_1 - \tilde{\mu}_2)^T \text{ [Since, it is a scalar]} \\ &= \left(\boldsymbol{w}^T\boldsymbol{\mu}_1 - \boldsymbol{w}^T\boldsymbol{\mu}_2\right)\left(\boldsymbol{w}^T\boldsymbol{\mu}_1 - \boldsymbol{w}^T\boldsymbol{\mu}_2\right)^T \\ &= \boldsymbol{w}^T \underbrace{(\tilde{\boldsymbol{\mu}}_1 - \tilde{\boldsymbol{\mu}}_2)(\tilde{\boldsymbol{\mu}}_1 - \tilde{\boldsymbol{\mu}}_2)^T}_{S_B} \boldsymbol{w}\end{aligned} \tag{5}$$

§ S_B which is a $d \times d$ matrix and has the form of covariance matrix is called the "between classes scatter matrix".

Linear Discriminant Analysis

$$J(\boldsymbol{w}) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2}$$

§ Let us play with the denominator.

$$\begin{aligned}
\tilde{S}_1^2 &= \sum_{i \in \mathcal{C}_1} (y_i - \tilde{\mu}_1)^2 = \sum_{i \in \mathcal{C}_1} (\boldsymbol{w}^T \boldsymbol{x}_i - \boldsymbol{w}^T \boldsymbol{\mu}_i)^2 \\
&= \sum_{i \in \mathcal{C}_1} (\boldsymbol{w}^T \boldsymbol{x}_i - \boldsymbol{w}^T \boldsymbol{\mu}_i)(\boldsymbol{w}^T \boldsymbol{x}_i - \boldsymbol{w}^T \boldsymbol{\mu}_i)^T \ \text{[scalar]} \\
&= \sum_{i \in \mathcal{C}_1} \boldsymbol{w}^T (\boldsymbol{x}_i - \boldsymbol{\mu}_i)(\boldsymbol{x}_i - \boldsymbol{\mu}_i)^T \boldsymbol{w} \\
&= \boldsymbol{w}^T \underbrace{\left(\sum_{i \in \mathcal{C}_1} (\boldsymbol{x}_i - \boldsymbol{\mu}_i)(\boldsymbol{x}_i - \boldsymbol{\mu}_i)^T \right)}_{S_1} \boldsymbol{w}
\end{aligned} \tag{6}$$

§ Similarly, $\tilde{S}_1^2 = \boldsymbol{w}^T S_2 \boldsymbol{w}$ comes from data in class 2.

Linear Discriminant Analysis

- Therefore, $\tilde{S}_1^2 + \tilde{S}_2^2 = \boldsymbol{w}^T \underbrace{(S_1 + S_2)}_{S_W} \boldsymbol{w}$
- S_W which is a $d \times d$ matrix and has the form of covariance matrix is called the "within class scatter matrix".

Linear Discriminant Analysis

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$$\text{So, } J(\boldsymbol{w}) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2} = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}} \tag{7}$$

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$$\text{So, } J(\boldsymbol{w}) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2} = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}} \tag{7}$$

- So, our optimization problem becomes,

$$\max_{\boldsymbol{w}} \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}} \tag{8}$$

- The solution strategy involves, maximizing $J(\boldsymbol{w})$ by taking the derivative w.r.t. $\boldsymbol{w}$ and setting it to 0.

LDA: Optimization

$$\frac{dJ(\boldsymbol{w})}{d\boldsymbol{w}} = \frac{\left(\frac{d}{d\boldsymbol{w}}\boldsymbol{w}^T S_B \boldsymbol{w}\right)\boldsymbol{w}^T S_W \boldsymbol{w} - \left(\frac{d}{d\boldsymbol{w}}\boldsymbol{w}^T S_W \boldsymbol{w}\right)\boldsymbol{w}^T S_B \boldsymbol{w}}{(\boldsymbol{w}^T S_W \boldsymbol{w})^2}$$

$$= \frac{(2S_B\boldsymbol{w})\boldsymbol{w}^T S_W \boldsymbol{w} - (2S_W\boldsymbol{w})\boldsymbol{w}^T S_B \boldsymbol{w}}{(\boldsymbol{w}^T S_W \boldsymbol{w})^2} = 0$$

§ So, we need to solve,

$$\begin{aligned}
(2S_B\boldsymbol{w})\boldsymbol{w}^T S_W \boldsymbol{w} - (2S_W\boldsymbol{w})\boldsymbol{w}^T S_B \boldsymbol{w} &= 0 \\
S_B\boldsymbol{w} - \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}} S_W\boldsymbol{w} &= 0 \\
S_B\boldsymbol{w} - \lambda S_W\boldsymbol{w} &= 0 \qquad \left[\text{Defining: } \lambda = \frac{\boldsymbol{w}^T S_B \boldsymbol{w}}{\boldsymbol{w}^T S_W \boldsymbol{w}}\right] \\
S_W^{-1} S_B\boldsymbol{w} - \lambda\boldsymbol{w} &= 0
\end{aligned}$$

LDA: Optimization

- So, we need to solve, $S_W^{-1} S_B \boldsymbol{w} = \lambda \boldsymbol{w}$.
- This almost looks like an eigenvalue equation. In fact, it is called a generalized eigenvalue problem.

LDA: Optimization

- So, we need to solve, $S_W^{-1} S_B \boldsymbol{w} = \lambda \boldsymbol{w}$.
- This almost looks like an eigenvalue equation. In fact, it is called a generalized eigenvalue problem.
- However, S_B being a product of $d \times 1$ and $1 \times d$ vectors, is rank deficient. (proof is skipped - revisit the concepts of column-space).
- However, S_B is always symmetric, positive semi-definite. And for such matrices one useful result (due to *Spectral Theorem* for symmetric matrices) is that $S_B = S_B^{1/2} S_B^{1/2}$.
- $S_B^{1/2}$ is computed as $S_B^{1/2} = Q\Lambda^{1/2}Q^T$, where the eigen decomposition of S_B gives, $S_B = Q\Lambda Q^T$. Then $\Lambda^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$.

$$S_W^{-1} S_B \boldsymbol{w} = S_W^{-1} S_B^{1/2} S_B^{1/2} \boldsymbol{w} = \lambda \boldsymbol{w}$$

$$S_B^{1/2} S_W^{-1} S_B^{1/2} \underbrace{S_B^{1/2} \boldsymbol{w}}_{\boldsymbol{v}} = \lambda \underbrace{S_B^{1/2} \boldsymbol{w}}_{\boldsymbol{v}}$$

$$S_B^{1/2} S_W^{-1} S_B^{1/2} \boldsymbol{v} = \lambda \boldsymbol{v}$$

- This is a regular eigenvalue problem for symmetric PSD matrix $S_W^{-1} S_B^{1/2} S_B^{1/2}$

LDA: Optimization

$$S_B^{1/2} S_W^{-1} S_B^{1/2} \boldsymbol{v} = \lambda \boldsymbol{v}$$

- We can find the solution then as, λ_k and $\boldsymbol{v}_k$ which in turn, would correspond to $\boldsymbol{w}_k = S_B^{-1/2} \boldsymbol{v}_k$
- Plugging the solution back in to the objective J, we can find that the eigenvector corresponding to the largest eigenvalue maximizes the objective. So, this is the desired direction.